

CSE422: Artificial intelligence

Assignment 04 [Fall 2025] v1.0

I highly recommended solving all these problems for a good preparation

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Probability

1. We construct a joint distribution table from three random variables: Device = {Laptop, Mobile}, User = {Student, Teacher}, Performance = {High, Low}

Device	User	Performance	Probability
Laptop	Student	High	0.12
Laptop	Student	Low	0.08
Laptop	Teacher	High	0.15
Laptop	Teacher	Low	0.05
Mobile	Student	High	0.2
Mobile	Student	Low	0.07
Mobile	Teacher	High	0.18
Mobile	Teacher	Low	0.15

- (a) Calculate the marginal probability of a user being a teacher. [Basically, calculate $P(\text{teacher})$.]
- (b) If you are given a mobile device, calculate the probability of it having a low performance. [Calculate $P(\text{Low} \mid \text{Mobile})$]
- (c) Are **User** and **Performance** conditionally independent of each other, given **Device = Mobile**?

2. A company uses an email spam filter. The filter correctly detects spam emails 90% of the time. It incorrectly marks a legitimate email as spam 4% of the time. In the company's inbox, 15% of all emails are spam. If the filter marks an email as spam, what is the probability that it is actually not spam?

Naive-Bayes Classifier

3. A national park uses sensors and ranger observations to classify whether an animal sighting belongs to a rare species.

Sighting ID	Time of Day	Location	Sound Detected?	Rare Species?
A01	Morning	Lakeside	Yes	Yes

A02	Night	Deep forrest	No	No
A03	Evening	Deep forrest	Yes	Yes
A04	Morning	Grassland	No	No
A05	Night	Lakeside	Yes	Yes
A06	Evening	Grassland	No	No
A07	Morning	Deep forrest	Yes	Yes
A08	Night	Grassland	Yes	No

Using the Naive Bayes classifier, predict whether the animal you saw last **evening, at lakeside, and heard barking** belongs to a rare species. [Hint: Basically, find out $P(\text{Rare} = \text{Yes} \mid \text{TimeOfDay} = \text{Evening}, \text{Location} = \text{Lakeside}, \text{SoundDetected} = \text{Yes})$ and $P(\text{Rare} = \text{No} \mid \text{TimeOfDay} = \text{Evening}, \text{Location} = \text{Lakeside}, \text{SoundDetected} = \text{Yes})$. Predict the one with higher value.]

4. Consider the following dataset, where x_1 and x_2 given y belong to a multinomial distribution:

x_1	x_2	y
0	1	0
1	0	0
1	1	1
0	1	0
0	0	1
1	1	2

- Using naïve bayes, calculate the most likely outcome of y given $x_1 = 1$ and $x_2 = 1$
- Consider that a new feature x_3 with continuous values, that is $x_3 \in \mathbb{R}$, is to be added to the dataset. How can the naive Bayes classifier be applied with x_3 .

Linear Regression and Logistic Regression

5. You are given the following dataset. You want to fit a linear regression model on this dataset with initial parameters $m = 2$ and $b = 1$.

x	y
1	5
3	12
5	15

- Calculate the **Mean Squared Error (MSE)** for this model on the dataset.
- Perform TWO iterations of **gradient descent** to update both m and b , assuming the learning rate = 0.05.

6. Consider the following regression dataset:

Sample	Hours studied (x1)	Hours slept (x2)	Test score (y)
1	1	3	5
2	2	2	4
3	0	4	1
4	3	1	9

You decide to use linear regression to solve this problem. For the weights, choose any value between [0.1, 0.9]. The learning rate must be between [0.01, 0.09].

- Calculate the **Mean Squared Error (MSE)** for this model on the dataset.
- Perform TWO iterations of **gradient descent** to update the parameters. Report the updated values.
- Using the updated values, find out the predicted test score for a student who studied 2.5 hours and slept 2 hours.

7. You want to fit a logistic regression model for the following classification dataset. You start with weight = 3 and bias = 1.

X	Y
0.2	0
0.4	0
0.8	1

- What type of classification is this? What would be a suitable loss function? [Hint: BCE]
- Calculate the amount of error using your chosen loss function.
- Perform two iterations of **gradient descent** to update the model parameters, assuming the learning rate = 0.1.

8. Consider the following classification dataset:

Sample	Cholesterol level (x1)	Exercise hour (x2)	Has Disease? (y)
1	6.5	3	0
2	6.2	0	1
3	5.0	4	0
4	7.8	1	1

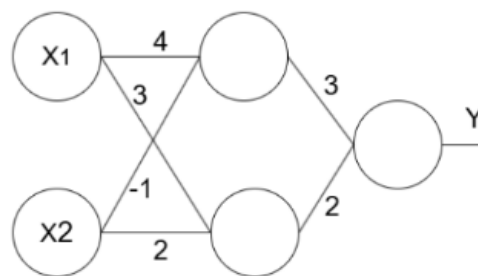
You decide to use logistic regression to solve this problem. For the weights, choose any value between [0.1, 0.9]. The learning rate must be between [0.01, 0.09].

- Calculate the error using **Binary Cross-Entropy (BCE)** for this model on the dataset.

- (e) Perform TWO iterations of **gradient descent** to update the parameters. Report the updated values.
- (f) Using the updated values, predict whether a person with a cholesterol level of 4.0 and 1 hour of exercise will have disease.

MLP (Neural Network)

9. Consider the following feed-forward neural network. Here the circles represent a perceptron (neuron), which is a weighted sum followed by an activation function. The hidden layer neurons use ReLU activation while the output layer uses sigmoid activation.



If $x_1 = 5$ and $x_2 = 1$, apply forward propagation to calculate the value of y .

10. Suppose we are training a Feed-forward Neural Network with ONE feature. Network structure is as follows:

Input feature, $x_1 = 1$

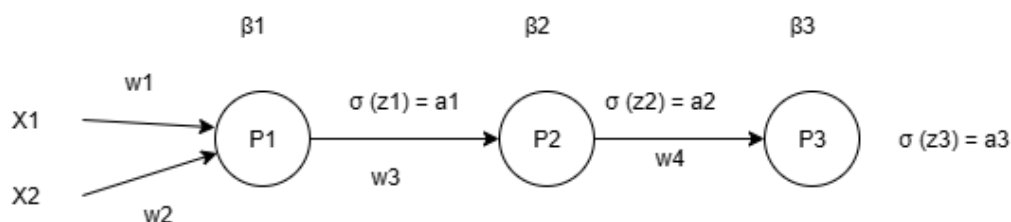
Hidden layer: 2 Neurons, weights are - 0.3 and 0.4, ReLU activation

Output layer: 1 neuron, weights are 1 and 2, Sigmoid activation

Loss function: Binary Cross Entropy

- (a) Draw the neural network.
- (b) Run the forward propagation and find the output.
- (c) If given output $y = 1$, calculate the error using the loss function.
- (d) Update all weights in the neural network using back propagation. You must show the derivatives. Take learning rate = 0.1.

11.



Suppose you are given the above architecture as an MLP for some inferential task. Your test set looks like $x_1 = 1$, $x_2 = 2$, and $y = 1$ (true label). Furthermore, the network assumes the weights and biases are initialized as $w_1 = 0.1$, $w_2 = 0.5$, $w_3 = 0.5$, $w_4 = 0.2$, $\beta_1 = 0.5$, $\beta_2 = 0.5$, and $\beta_3 = 0.5$, respectively

as part of its initial hypothesis. Moreover, P1, P2, and P3 are the three perceptrons whose non-linear outputs are a_1 , a_2 , and y' , respectively. The Loss function is Binary Cross Entropy (BCE).

- Write the equation of the loss function.
- Perform one pass of forward propagation and determine the predicted output for the given test set.
- For one pass of backward propagation, first derive the expression for $\partial L / \partial w_4$. Then use the chain rule to derive the expression for $\partial L / \partial w_1$. Finally, calculate the quantitative values.

Decision Tree

12. A bank wants to predict if a customer will **default** on a loan (Yes, No) based on their **credit score** (Low, Medium, High) and previous **loan history** (Good, Bad). Altogether, the dataset has records of **100 people**, with the following distribution:

- 45 people with Low credit score: 15 default, 25 do not default.
- 30 people with Medium credit score: 10 default, 20 do not default.
- 20 people with High credit score: 5 default, 15 do not default.
- 50 people with Good loan history: 5 default, 45 do not default.
- 50 people with Bad loan history: 25 default, 25 do not default.

Questions:

- Calculate the entropy of the dataset. [Hint: entropy of a dataset = $H(\text{target}) = H(\text{default})$ in this case]
- What is the information gain if we split the dataset based on **credit score**? [Hint: find information gain (default, credit score) = $H(\text{default}) - \text{Remainder}(\text{credit score})$]
- What is the information gain if we split the dataset based on **load history**?

13. Decision tree construction. For the following dataset, construct the full decision tree using the ID3 algorithm shown in class.

Attributes:

- Age Group: Young (Y), Middle-aged (M), Senior (S)
- Employment Status: Employed (E), Unemployed (U)
- Interest in Community Events: Low (L), Medium (M), High (H)

Target Variable: Participation (Yes, No)

Age Group	Employment Status	Interest in Community Events	Participation
Young	Employed	High	Yes
Young	Employed	Medium	Yes
Young	Unemployed	Low	No
Young	Unemployed	High	Yes

Middle-aged	Employed	Medium	Yes
Middle-aged	Employed	Low	No
Middle-aged	Unemployed	Medium	No
Senior	Employed	High	Yes
Senior	Unemployed	Low	No
Senior	Unemployed	Medium	No
Senior	Unemployed	Low	No
Young	Employed	Low	No
Middle-aged	Employed	High	Yes
Senior	Unemployed	High	Yes

Hint:

- Before attempting this problem, make sure you understand the problem shown in slide thoroughly.
- First calculate Gain(Participation, Age Group), Gain(Participation, Employment), Gain(Participation, Interest). Choose the attribute with the highest gain.
- Suppose Employment Status has the highest gain. Choose this as the root of the tree. Since Employment Status has two classes (employed, unemployed), root will have two children. We now need to choose the next attribute.
- Calculate Gain(Age Group, employed) and Gain(Interest, employed). Choose the one with the highest gain.
- Similarly, calculate Gain(Age Group, unemployed) and Gain(Interest, unemployed). Choose the one with the highest gain.]
- Continue in this way until you reach the leaf nodes (yes or no).

14. Using the decision tree you constructed in Q2, predict whether the following customers would default the loan:

- Customer 1: middle-aged, unemployed, low interest
- Customer 2: Senior, employed, medium interest

15. The following table shows a set of conditions and actions performed by a cleaning bot. The agent uses the ID3 algorithm (Decision Tree algorithm) to make its decisions.

Current Room	Time of Day	Battery Level	Action
Not Dirty	Morning	High	Move to another room
Dirty	Morning	High	Clean room
Not Dirty	Afternoon	Medium	Move to another room
Dirty	Evening	Low	Return to base
Dirty	Afternoon	Medium	Clean room
Not Dirty	Morning	Low	Return to base
Not Dirty	Afternoon	Medium	Move to another room
Dirty	Evening	Low	Return to base

Not Dirty	Evening	High	Move to another room
Dirty	Morning	Low	Return to base

- (a) Given $H(\text{Action}) = 1.522$ and $I(\text{Action}, \text{Time of Day}) = 0.371$, calculate the value of $H(\text{Action} | \text{Time of Day} = \text{Morning})$.
- (b) Calculate $I(\text{Action}, \text{Battery Level})$, given that we have computed $I(\text{Action}, \text{Current Room}) = 0.675$. Draw the first level of the Decision Tree including any pure leaf nodes.

16. The following information is collected from a dataset including three **types of plants** (Shrub, Flower and Tree) with three **colors** (Red, Yellow, Blue).

Overall Type Probabilities: $P(\text{Shrub}) = P(\text{Flower}) = P(\text{Tree}) = 1/3$.

Color Distributions for Each Type:

- For Shrubs: $P(\text{Red} | \text{Shrub}) = 1/4$, $P(\text{Yellow} | \text{Shrub}) = 1/2$, $P(\text{Blue} | \text{Shrub}) = 1/4$.
- For Flowers: $P(\text{Red} | \text{Flower}) = 1/2$, $P(\text{Yellow} | \text{Flower}) = 0$, $P(\text{Blue} | \text{Flower}) = 1/2$.
- For Trees: $P(\text{Red} | \text{Tree}) = 0$, $P(\text{Yellow} | \text{Tree}) = 0$, $P(\text{Blue} | \text{Tree}) = 1$

Consider **plant type** to be the class variable (i.e. target) and **colors** to be an attribute.

- (a) Calculate the Entropy of the dataset. [Hint: Find $H(\text{Type})$ since Type is the target in this dataset.]
- (b) Calculate the Conditional Entropy $H(\text{Type} | \text{Color})$ for attribute **Color** with respect to the type.
- (c) Calculate the Information Gain, $I(\text{Type}, \text{Color})$ for attribute **Color** with respect to predicting the type.